

Education

Carnegie Mellon University

Master's in Robotics, GPA: 4.24 / 4.00

Aug 2025 - Aug 2027 (Expected)

Pittsburgh, PA

BITS Pilani

B.E. Computer Science + M.Sc. Economics (Dual Major), GPA: 8.67 / 10.00

Aug 2018 - May 2023

Pilani, India

- **Thesis Advisor:** Prof. R. Venkatesh Babu, Vision and AI Lab, IISc Bangalore

Experience

Samsung R&D Institute India

Nov 2023 – May 2025

Senior Software Engineer

Bengaluru, India

- Developed a **multilingual BERT-based LLM safety classifier** across **12+** locales, achieving **95%** accuracy and improving robustness against unsafe, adversarial, and prompt-injection inputs.
- Designed and implemented an **Android system service** to enable on-device language & vision models across flagship devices, used in notes app, camera, keyboard; reduced inference latency by **~5s** and memory usage by **~1GB**.

PrivateBlok (acquired by KPMG)

Feb 2023 – Nov 2023

Data Science Intern

Bengaluru, India

- Built a GPT-3.5 Turbo **financial QA chatbot** for **10K+** private companies in India, integrating structured company data with **RAG**, optimized vector **retrieval** and custom re-ranking.
- Improved answer grounding by vectorizing a **temporal knowledge graph** and optimizing the GPT-LangChain pipeline, reducing *hallucinations* and improving context retrieval for financial QA.

Projects

3D Hand-Object Interaction Generation | Submitted to ECCV 2026, Accepted in 1st PhysHuman @ CVPR 2026 workshop

- Built a **masked auto-regressive diffusion** framework for **text-conditioned 3D hand-object interaction generation**, enabling smooth variable-length motion generation, composite sequencing, completion, and infilling.
- Achieved **SOTA** generation quality on ARCTIC and GRAB, reaching Top-3 retrieval accuracy of **0.986** on ARCTIC and reducing FID to **0.188**, outperforming OpenHOI, Text2HOI, DiffH2O, and LatentHOI.

FPS-Bench: A Benchmark for Video Understanding at High Frame Rates

- Developed FPS-Bench, a **high-frame-rate video QA benchmark** with **minFPS** annotations; evaluated open/closed SOTA VLMs and exposed a major gap in fine-grained temporal reasoning.
- Curated **1K** questions from 554 videos across 9 temporal categories, requiring ≥ 4 FPS and **6.67 average minFPS**; SOTA VLMs reached only **31.8%** accuracy vs. **72.2% human performance**.

Long-tailed Image Generation using StyleGANs | Undergraduate Thesis at Vision & AI Lab, IISc Bangalore

- Analyzed and implemented state-of-the-art GAN conditioning and regularization techniques to address challenges in **long-tailed** image generation, prevalent in real-world datasets.
- Mitigated mode-collapse and class-confusion for tail classes by de-correlating w latent space, setting a new benchmark on long-tailed datasets like ImageNet-LT and iNaturalist2019 with average FID improvement of **19%**.

A Generalized Multimodal Deep Learning Model for Early Crop Yield Prediction | Year-long study-oriented project

- Proposed and developed a generalized multi-modal deep-learning model, **CropYieldNet**, for crop yield prediction with **limited** high-resolution satellite data, achieving superior **generalizing capability** compared to existing models.
- Improved model performance up to **7.8%** by modeling the depth variant information of soil data and achieved the best in-season RMSE of **11.328** for India and **5.99** for U.S. for corn and soybean crop yield.

Publications

- *FPSBench: A Benchmark for Video Understanding at High Frame Rates* [↗](#)
Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (**CVPR 2026**)
- *NoisyTwins: Class-Consistent and Diverse Image Generation through StyleGANs* [↗](#)
Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (**CVPR 2023**)
- *A Generalized Multimodal Deep Learning Model for Early Crop Yield Prediction* [↗](#)
Published in 2022 IEEE International Conference on Big Data (Big Data) (**Big Data 2022**)

Technical Skills

Languages: Python, Java, C++, Kotlin, Git, Linux, $\LaTeX$

ML / AI: PyTorch, TensorFlow, Diffusers, LangChain, OpenCV, scikit-learn, NumPy, Pandas, Keras

Robotics / Simulation: Isaac Sim, ManiSkill, Newton Physics Engine, MuJoCo